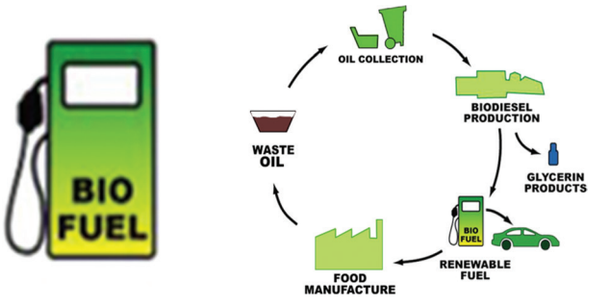


Figure 3.2.1.4 Bio Diesel Production Process

3.2.2 Gaseous Fuels

The gaseous fuels are readily mix with atmospheric air without delay and it is inducted in engine. The following gaseous fuels are presently in use.

3.2.2.1 Liquified Petroleum Gas (LPG)

Liquified petroleum gas (LPG) also called as Propane is a by product of natural gas processing and crude oil refining. LPG is widely used as a fuel for domestic cooking and heating and now it is also a popular alternative fuel for vehicles. It is stored under pressure (100psi or 680 atm) inside a special tank and is a colorless, odorless liquid. As pressure is released, the liquid propane vaporizes and turns into gas that is used in combustion.An odorant, ethyl mercaptan, is added for leak detection.Propane has a high-octane rating, making it an excellent choice for spark-ignited internal combustion engines. It provides uniform homogenous mixture for all cylinders. The carbon content in LPG is less than petrol and hence, LPG vehicles can produce lower amounts of harmful air pollutants and greenhouse gases, CO_2. The operating cost of the vehicle is reduced by 50%. The use of LPG enhances engine life. Refer Figure 3.2.2.1.



Figure 3.2.2.1 Liquified Petroleum Gas

3.2.2.2 Liquified Natural Gas (LNG)

Liquified natural gas (LNG), is natural gas in its liquid form. LNG is produced by purifying natural gas and super-cooling it to -161°C to turn it into a liquid. During the process known as liquefication, natural gas is cooled below its boiling point, removing most of the extraneous compounds found in the fuel. The remaining natural gas is primarily methane (98%) with small amounts of other hydrocarbons.

The specific gravity of LNG is higher than CNG. The calorific value of LNG is 48MJ/kg and its octane value are 110. Because of LNG's relatively high production cost as well as the need to store it in expensive cryogenic tanks, the commercial applications of LNG has been limited. Refer Figure 3.2.2.2.

3.2.2.3 Compressed Natural Gas (CNG)

Natural gas is primarily extracted from gas and oil wells. Natural gas is an odorless and it is a mixture of hydrocarbon, mainly 95% of methane and 5% of other components like butane, propane, ethane,

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